Continuous-Time Portfolio Optimization: The Merton Problem

Abstract

This paper studies the classical Merton portfolio problem in continuous time. We consider an investor who dynamically allocates wealth between a risk-free asset and a risky asset in order to maximize expected utility of terminal wealth. We derive the Hamilton–Jacobi–Bellman (HJB) equation and obtain closed-form expressions for both the value function and the optimal portfolio rule under constant relative risk aversion (CRRA) utility. The optimal fraction of wealth invested in the risky asset is constant over time and depends on the market risk premium, asset volatility, and the investor's coefficient of relative risk aversion. We then use historical data and Monte Carlo simulation to illustrate the model, examine the distribution of terminal wealth, and compare the optimal rule with alternative allocation strategies. Finally, we analyze the welfare consequences of expected-return estimation error using certainty equivalents and evaluate a constrained version of the Merton rule in a historical backtest. Together, the analytical and numerical results provide a unified introduction to optimal portfolio choice in continuous time while highlighting the practical importance of parameter uncertainty and portfolio constraints.

1 Introduction

Financial markets offer opportunities for growth while exposing investors to uncertainty and loss. Portfolio choice therefore requires a systematic framework for allocating capital across assets with different expected returns and risk characteristics. A useful model should identify the trade-off between return and risk, describe how that trade-off changes with investor preferences, and provide an implementable investment rule.

Modern portfolio theory began with Markowitz's mean–variance framework [1], which formalized portfolio selection by evaluating assets jointly rather than in isolation. Sharpe subsequently developed the Capital Asset Pricing Model (CAPM), linking expected returns to systematic market risk [3]. Modigliani and Miller also made foundational contributions to the theory of corporate finance and market valuation [4]. Markowitz, Sharpe, and Merton Miller received the 1990 Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel for this body of work [7, 8].

Merton extended static portfolio theory to a dynamic, continuous-time setting [2]. In the classical model, risky asset prices follow geometric Brownian motion and investor wealth evolves according to a controlled stochastic differential equation. Dynamic programming and Itô calculus then yield explicit portfolio rules for several utility specifications. A central implication is a two-fund separation result: the investor combines a risk-free asset with a risky fund, and the relative allocation is determined by market opportunities and risk preferences.

This paper develops and illustrates the Merton problem under CRRA utility. Section 2 introduces the financial market, wealth dynamics, and optimization objective. Section 3 derives the HJB equation, solves for the optimal risky allocation, and presents comparative statics. Section 4 describes the data calibration and simulation framework. Section 5 studies welfare losses caused by expected-return estimation error. Section 6 presents a historical backtest of a constrained Merton strategy. Section 7 summarizes the main findings, limitations, and possible extensions.

2 Model Formulation

We consider a continuous-time financial market over a finite horizon $[0, T]$ with two traded assets.

(a) The risk-free asset B_t satisfies
$$dB_t = rB_t\,dt,$$
where r is the constant continuously compounded risk-free rate.

(b) The risky asset S_t follows geometric Brownian motion:
$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t,$$
where $\mu \in \mathbb{R}$ is the expected rate of return, $\sigma > 0$ is the volatility, and $\{W_t\}_{t\geq 0}$ is a standard Brownian motion.

Let X_t denote investor wealth at time t. Throughout the paper, π_t denotes the *fraction of wealth* invested in the risky asset. Thus, $\pi_t X_t$ is invested in the risky asset and $(1 - \pi_t)X_t$ is invested in the risk-free asset. Values $\pi_t > 1$ represent leveraged risky positions, whereas $\pi_t < 0$ represent short positions in the risky asset.

2.1 Wealth Dynamics

If the investor consumes at rate $c_t \geq 0$, the self-financing wealth process satisfies
$$dX_t = \big[X_t\big(r + \pi_t(\mu - r)\big) - c_t\big]\,dt + \pi_t\sigma X_t\,dW_t.$$

The analysis below assumes no intermediate consumption, so $c_t \equiv 0$ and
$$dX_t = X_t\big[r + \pi_t(\mu - r)\big]dt + \pi_t\sigma X_t\,dW_t. \tag{1}$$

We restrict attention to admissible portfolio processes for which the stochastic differential equation is well defined and expected utility is finite.

2.2 Utility and Value Function

The investor has CRRA utility over terminal wealth:
$$U(x) = \frac{x^{1-\gamma}}{1-\gamma}, \qquad \gamma > 0, \quad \gamma \neq 1, \tag{2}$$
where γ is the coefficient of relative risk aversion. The limiting case $\gamma = 1$ corresponds to logarithmic utility, $U(x) = \ln x$, up to an irrelevant additive constant.

The investor chooses an admissible portfolio process $\pi = \{\pi_t\}_{0\leq t\leq T}$ to maximize expected utility of terminal wealth:
$$\sup_\pi \mathbb{E}\big[U(X_T)\big].$$
The associated value function is
$$V(t, x) = \sup_\pi \mathbb{E}_{t,x}\big[U(X_T)\big], \tag{3}$$
where $\mathbb{E}_{t,x}$ denotes conditional expectation given $X_t = x$.

3 Analytical Results

We derive the HJB equation and solve for the optimal portfolio rule following Merton [2]; standard stochastic-calculus references include [9, 10].

3.1 Derivation of the HJB Equation

The dynamic programming principle states that an optimal strategy over $[t, T]$ must remain optimal after any intermediate date. Over a short interval $[t, t + \Delta t]$,

$$V(t, x) = \sup_{\pi} \mathbb{E}_{t,x} \left[V(t + \Delta t, X_{t+\Delta t}) \right].$$

Applying Itô's lemma and taking the limit as $\Delta t \to 0$ converts this global optimization problem into a local one.

Itô's lemma. Let

$$dX_t = a(t, X_t)\, dt + b(t, X_t)\, dW_t$$

and let $g(t, x)$ be once continuously differentiable in t and twice continuously differentiable in x. Then

$$dg(t, X_t) = \left(g_t + a g_x + \frac{1}{2} b^2 g_{xx} \right) dt + b g_x\, dW_t.$$

Applying Itô's lemma to $V(t, X_t)$ and using (1) gives

$$\begin{aligned} dV(t, X_t) = &\Big[V_t + \big(r + \pi_t(\mu - r)\big) X_t V_x + \frac{1}{2} \pi_t^2 \sigma^2 X_t^2 V_{xx} \Big] dt \\ &+ \pi_t \sigma X_t V_x\, dW_t. \end{aligned}$$

The conditional expectation of the stochastic-integral term is zero under the usual integrability conditions. Therefore, the HJB equation is

$$V_t(t, x) + \sup_{\pi \in \mathbb{R}} \left\{ [r + \pi(\mu - r)] x V_x(t, x) + \frac{1}{2} \pi^2 \sigma^2 x^2 V_{xx}(t, x) \right\} = 0, \tag{4}$$

with terminal condition

$$V(T, x) = U(x) = \frac{x^{1-\gamma}}{1 - \gamma}. \tag{5}$$

3.2 Computation of the Optimizer

For fixed (t, x), define the terms in (4) that depend on π by

$$F(\pi) = \pi(\mu - r) x V_x + \frac{1}{2} \pi^2 \sigma^2 x^2 V_{xx}.$$

The first-order condition is

$$F'(\pi) = (\mu - r) x V_x + \pi \sigma^2 x^2 V_{xx} = 0,$$

which yields

$$\pi^*(t, x) = -\frac{(\mu - r) V_x(t, x)}{\sigma^2 x V_{xx}(t, x)}. \tag{6}$$

Because

$$F''(\pi) = \sigma^2 x^2 V_{xx},$$

the critical point is a maximum whenever the value function is concave in wealth, so that $V_{xx} < 0$.

Substituting (6) into the HJB equation gives the reduced equation

$$V_t + rxV_x - \frac{(\mu - r)^2}{2\sigma^2}\frac{V_x^2}{V_{xx}} = 0, \qquad V(T,x) = \frac{x^{1-\gamma}}{1-\gamma}. \tag{7}$$

3.3 Homothetic Form and Closed-Form Solution

CRRA utility is homogeneous of degree $1-\gamma$:

$$U(\lambda x) = \lambda^{1-\gamma}U(x), \qquad \lambda > 0.$$

Because the wealth equation is linear in wealth, the value function inherits the same scaling property. We therefore use the ansatz

$$V(t,x) = \frac{A(t)}{1-\gamma}x^{1-\gamma}, \qquad A(T) = 1. \tag{8}$$

Its derivatives are

$$V_t = \frac{A'(t)}{1-\gamma}x^{1-\gamma}, \qquad V_x = A(t)x^{-\gamma}, \qquad V_{xx} = -\gamma A(t)x^{-\gamma-1}.$$

Substituting these derivatives into (6) gives the constant optimal risky fraction

$$\boxed{\pi^* = \frac{\mu - r}{\gamma\sigma^2}}. \tag{9}$$

Consequently, the dollar amount invested in the risky asset is $\pi^* X_t$.

Substituting the ansatz into (7) yields

$$A'(t) + (1-\gamma)\left[r + \frac{(\mu - r)^2}{2\gamma\sigma^2}\right]A(t) = 0.$$

Using $A(T) = 1$, we obtain

$$A(t) = \exp\left\{(1-\gamma)\left[r + \frac{(\mu - r)^2}{2\gamma\sigma^2}\right](T-t)\right\}.$$

Therefore, the closed-form value function is

$$\boxed{V(t,x) = \frac{x^{1-\gamma}}{1-\gamma}\exp\left\{(1-\gamma)\left[r + \frac{(\mu - r)^2}{2\gamma\sigma^2}\right](T-t)\right\}.} \tag{10}$$

3.4 Comparative Statics

The Merton rule in (9) directly shows how the optimal risky allocation depends on model parameters.

Risk aversion. If $\mu > r$, then

$$\frac{\partial \pi^*}{\partial \gamma} = -\frac{\mu - r}{\gamma^2 \sigma^2} < 0.$$

More risk-averse investors allocate a smaller fraction of wealth to the risky asset.

Volatility. If $\mu > r$, then

$$\frac{\partial \pi^*}{\partial \sigma} = -\frac{2(\mu - r)}{\gamma \sigma^3} < 0.$$

Higher volatility reduces the optimal risky allocation.

Risk premium.

$$\frac{\partial \pi^*}{\partial (\mu - r)} = \frac{1}{\gamma \sigma^2} > 0.$$

A larger expected excess return increases the optimal risky allocation.

Investment horizon. Under the classical assumptions of constant investment opportunities and terminal-wealth CRRA utility, π^* does not depend on t or T. The optimal risky fraction is therefore constant over time. The horizon still affects the distribution of terminal wealth and the value function.

4 Data Analysis and Simulation

4.1 Data Collection

We use the SPDR S&P 500 ETF Trust (SPY) as the risky asset in the calibration exercise. SPY is a broad and liquid proxy for the U.S. large-cap equity market and has an extensive price history.

Historical daily closing prices were collected using Microsoft Excel's built-in stock-history function:

```
=STOCKHISTORY("SPY", DATE(2026,1,1), DATE(2026,4,14)).
```

The calibration sample spans January 1, 2026, through April 14, 2026.

The risk-free rate is approximated by the three-month U.S. Treasury bill rate (DTB3) from the Federal Reserve Economic Data database:

`https://fred.stlouisfed.org/series/DTB3`.

We use the average DTB3 observation over the same sample period and convert percentage observations to decimal form before applying the portfolio formula.

4.2 Data Cleaning

For the SPY series, observations without a valid closing price are removed. Such gaps generally correspond to weekends, holidays, or other non-trading days. The Treasury bill series is restricted to the same date range, and missing observations are excluded before computing the sample average. Dates are aligned so that the parameter estimates use internally consistent inputs.

4.3 Parameter Estimation and the Optimal Strategy

Let the daily log return be

$$R_t = \ln\left(\frac{S_t}{S_{t-1}}\right).$$

Using 252 trading days per year, define the annualized mean log return and volatility by

$$\widehat{m} = 252\,\overline{R}, \qquad \widehat{\sigma} = \sqrt{252}\,s_R,$$

where $\overline{R}$ and s_R are the sample mean and sample standard deviation of daily log returns. Under geometric Brownian motion,

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\, dW_t,$$

so the corresponding estimate of the price-process drift is

$$\widehat{\mu} = \widehat{m} + \frac{1}{2}\widehat{\sigma}^{\,2}.$$

The risk-free rate is estimated as

$$\widehat{r} = \mathrm{avg}(DTB3).$$

Using the reported sample statistics gives

- annualized mean log return: $\widehat{m} = 0.059862 \approx 5.99\%$;
- annualized GBM drift: $\widehat{\mu} = 0.071001 \approx 7.10\%$;
- annualized volatility: $\widehat{\sigma} = 0.149260 \approx 14.93\%$;
- annualized risk-free rate: $\widehat{r} = 0.035968 \approx 3.60\%$.

For logarithmic utility, corresponding to $\gamma = 1$, the Merton rule becomes

$$\pi^* = \frac{\mu - r}{\sigma^2}.$$

Using the internally consistent GBM estimates,

$$\widehat{\pi}^* = \frac{0.0710 - 0.0360}{0.1493^2} \approx 1.5725.$$

Thus, the unconstrained model prescribes investing approximately 157.25% of wealth in SPY and −57.25% in the risk-free asset. Economically, the negative risk-free weight means borrowing an amount equal to 57.25% of current wealth at the risk-free rate. This interpretation relies on frictionless borrowing; a no-leverage constraint would cap the risky weight at 100%.

Because the calibration window is short, the expected-return estimate is likely to be noisy. The large change in the recommended allocation also illustrates the Merton rule's sensitivity to seemingly modest differences in return-estimation conventions. The numerical allocation should therefore be viewed as a model illustration rather than an investment recommendation.

4.4 Geometric Brownian Motion Simulation

The risky asset follows

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t.$$

Applying Itô's lemma to $\ln S_t$ gives

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma \, dW_t.$$

Over a time step Δt,

$$W_{t+\Delta t} - W_t = \sqrt{\Delta t}\, Z_t, \qquad Z_t \sim N(0,1),$$

so the exact discrete-time simulation formula is

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_t\right]. \tag{11}$$

4.5 Wealth Dynamics Under a Constant Portfolio Weight

For a constant risky fraction π, wealth satisfies

$$dX_t = X_t\big[\pi\mu + (1-\pi)r\big]dt + X_t\pi\sigma \, dW_t.$$

Its exact time-step update is

$$X_{t+\Delta t} = X_t \exp\left\{\left[\pi\mu + (1-\pi)r - \frac{1}{2}\pi^2\sigma^2\right]\Delta t + \pi\sigma\sqrt{\Delta t}\, Z_t\right\}. \tag{12}$$

The terminal observation from each simulated path forms one draw from the terminal-wealth distribution.

When realized simple returns $R_{t+\Delta t}$ are used instead of simulated Brownian shocks, the corresponding portfolio-accounting update is

$$X_{t+\Delta t} = X_t \left[1 + \pi R_{t+\Delta t} + (1-\pi)r\Delta t\right]. \tag{13}$$

Equation (13) is a return aggregation formula and should be distinguished from the exact lognormal update in (12).

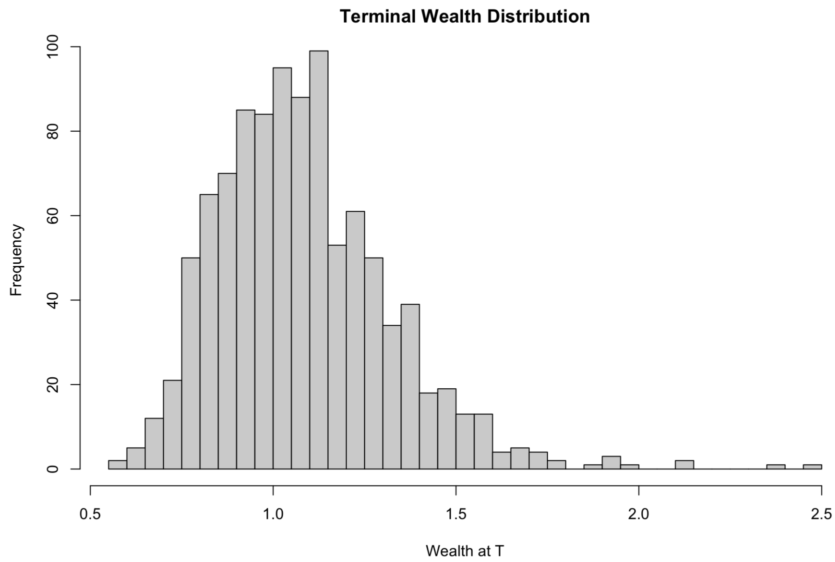

Figure 1: Simulated distribution of terminal wealth across Monte Carlo paths.

4.6 Comparison with Alternative Allocation Strategies

The optimal allocation is compared with a fixed 60% risky allocation and an all-cash strategy. These benchmarks illustrate the trade-off between expected growth and dispersion in terminal wealth.

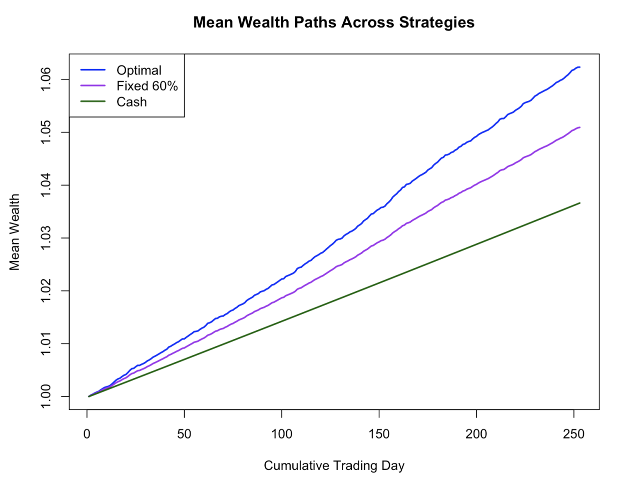

Figure 2: Mean wealth paths under the Merton, fixed 60% risky, and all-cash strategies. In the reported simulation, the Merton allocation has the fastest average growth, the fixed allocation has intermediate growth, and the cash strategy grows at the risk-free rate.

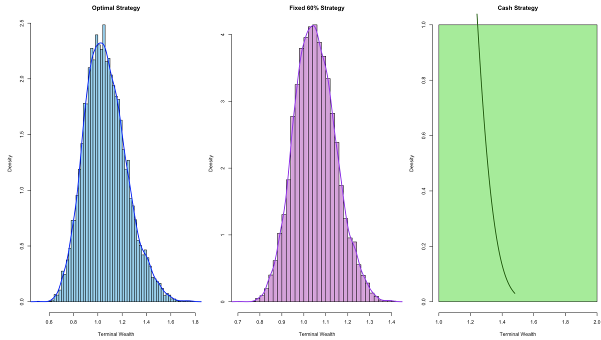

Figure 3: Simulated terminal-wealth distributions under the three strategies. The more aggressive allocations produce greater average terminal wealth together with wider dispersion.

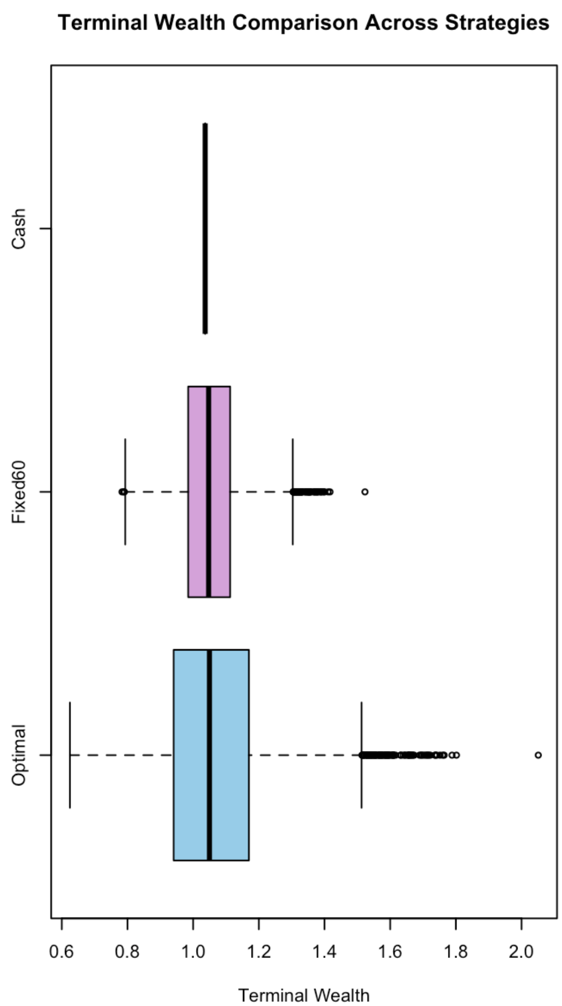

Figure 4: Comparison of terminal-wealth variability across strategies. Riskier allocations exhibit greater dispersion, whereas the cash strategy is comparatively stable.

5 Welfare and Certainty-Equivalent Sensitivity Analysis

5.1 Motivation

The Merton rule treats μ, r, and σ as known constants. In practice, these quantities must be estimated, and estimation error can materially affect the recommended portfolio. Expected returns are especially difficult to estimate precisely. We therefore quantify the welfare consequences of misspecifying μ while treating r and σ as known.

5.2 Certainty-Equivalent Framework

A certainty equivalent is the guaranteed amount of terminal wealth that gives the investor the same utility as a risky strategy. It translates expected utility into an economically interpretable wealth measure. For a risk-averse investor, the certainty equivalent is generally below expected terminal wealth because it accounts for uncertainty rather than only the mean outcome.

5.3 Expected Utility Under a Constant Portfolio Weight

For constant π, the wealth process has the solution

$$X_T = X_0 \exp\left\{\left[r + \pi(\mu - r) - \frac{1}{2}\pi^2\sigma^2\right] T + \pi\sigma W_T\right\}.$$

Under CRRA utility,

$$\mathbb{E}[U(X_T)] = \frac{X_0^{1-\gamma}}{1-\gamma} \exp\left\{(1-\gamma)g(\pi)T\right\},$$

where

$$g(\pi) = r + \pi(\mu - r) - \frac{1}{2}\gamma\pi^2\sigma^2. \tag{14}$$

Maximizing expected utility is therefore equivalent to maximizing $g(\pi)$. The first-order condition recovers

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2}.$$

Define the certainty equivalent $CE(\pi)$ by

$$U\big(CE(\pi)\big) = \mathbb{E}[U(X_T)].$$

Using the expression above,

$$CE(\pi) = X_0 \exp\big(g(\pi)T\big). \tag{15}$$

This formula also extends continuously to logarithmic utility.

5.4 Expected-Return Misspecification

Suppose the investor uses an estimated expected return

$$\widehat{\mu} = \mu + \varepsilon_\mu,$$

where ε_μ is the estimation error. The investor then chooses

$$\begin{aligned}\widehat{\pi} &= \frac{\widehat{\mu} - r}{\gamma\sigma^2} \\ &= \frac{(\mu - r) + \varepsilon_\mu}{\gamma\sigma^2} \\ &= \pi^* + \frac{\varepsilon_\mu}{\gamma\sigma^2}.\end{aligned}$$

Let

$$\delta = \widehat{\pi} - \pi^* = \frac{\varepsilon_\mu}{\gamma\sigma^2}.$$

Because $g(\pi)$ is a concave quadratic centered at π^*,

$$g(\pi^*) - g(\widehat{\pi}) = \frac{1}{2}\gamma\sigma^2\delta^2 \tag{16}$$

$$= \frac{\varepsilon_\mu^2}{2\gamma\sigma^2}. \tag{17}$$

Thus, the growth-rate loss is second order in the expected-return estimation error.

5.5 Welfare Loss

Combining (15) and (17), the ratio of certainty equivalents is

$$\frac{CE(\widehat{\pi})}{CE(\pi^*)} = \exp\left(-\frac{\varepsilon_\mu^2}{2\gamma\sigma^2}T\right). \tag{18}$$

The fractional welfare loss is therefore

$$\mathcal{L} = 1 - \frac{CE(\widehat{\pi})}{CE(\pi^*)} = 1 - \exp\left(-\frac{\varepsilon_\mu^2}{2\gamma\sigma^2}T\right). \tag{19}$$

For small losses, $1 - e^{-x} \approx x$, so

$$\mathcal{L} \approx \frac{\varepsilon_\mu^2}{2\gamma\sigma^2}T. \tag{20}$$

5.6 Interpretation

Equation (19) yields several insights. First, welfare loss increases quadratically with expected-return estimation error, so large errors become costly quickly. Second, the loss accumulates with the investment horizon. Third, for a fixed expected-return error, greater risk aversion reduces the loss because the investor takes a smaller risky position. Fourth, the same error is more damaging when true volatility is low because the Merton rule responds more aggressively to a given estimated risk premium.

This calculation examines misspecification of μ only. It does not establish that volatility estimation is generally less important; assessing errors in σ requires a separate derivation because volatility enters both the optimal weight and the distribution of wealth.

6 Empirical Verification

6.1 Assets and Sample

The empirical exercise uses the following assets:

- **Risky asset:** SPY, using dividend-adjusted closing prices as a proxy for the U.S. equity market.
- **Risk-free asset:** a fixed annual rate of 4%, approximated at the monthly frequency by $0.04/12 \approx 0.0033$.

The sample runs from February 1993 through December 2024. Daily adjusted closing prices are downloaded with `yfinance` and converted to monthly returns using end-of-month observations. Equivalently, daily simple returns within each month may be compounded as

$$R_t^{(m)} = \prod_{d \in t}(1 + R_d) - 1.$$

6.2 Rolling Parameter Estimation

At each month t, expected return and volatility are estimated from the preceding 12 monthly observations. If $\overline{R}_t$ and s_t denote the rolling monthly sample mean and standard deviation, the annualized estimates are

$$\widehat{\mu}_t = 12\overline{R}_t, \qquad \widehat{\sigma}_t = \sqrt{12}\, s_t.$$

The unconstrained Merton weight is

$$\widetilde{w}_t = \frac{\widehat{\mu}_t - r}{\gamma \widehat{\sigma}_t^2}.$$

Because the backtest does not allow leverage or short selling, the implemented weight is clipped to the interval $[0, 1]$:

$$w_t^* = \min\left\{1, \max\left\{0, \widetilde{w}_t\right\}\right\}. \tag{21}$$

6.3 Parameter Choices

The backtest uses

- risk aversion $\gamma = 0.5$;
- a 12-month rolling estimation window;
- monthly rebalancing; and
- a no-leverage, no-short-selling constraint.

The relatively low value of γ permits an aggressive risky allocation before the constraint is applied. The one-year window is intended to emphasize recent market conditions, although using only 12 monthly observations also makes the estimates noisy.

6.4 Portfolio Return

At month t, the portfolio return is

$$R_t^p = w_t^* R_t^{\text{risky}} + (1 - w_t^*) R_t^{\text{rf}}. \tag{22}$$

6.5 Benchmark Strategies

We compare the constrained Merton strategy with three benchmarks:

- **100% SPY:** passive long-only exposure to the equity market;
- **100% risk-free asset:** the low-risk baseline; and
- **50/50 portfolio:** an equal-weighted allocation rebalanced monthly.

6.6 Results

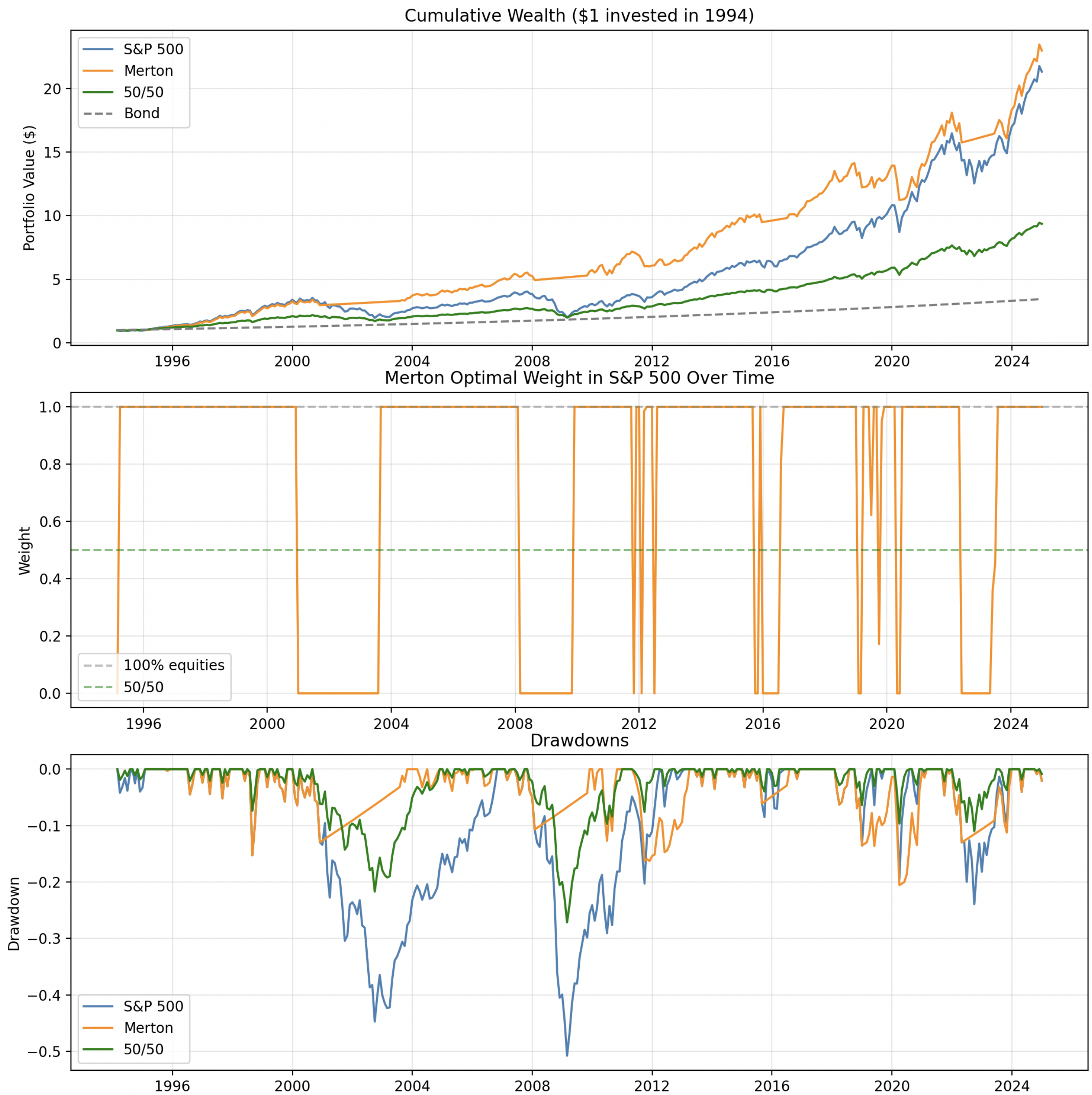

Figure 5: Historical cumulative performance of the constrained Merton strategy and benchmark portfolios.

In the reported backtest, the constrained Merton strategy achieves strong cumulative performance relative to the benchmark portfolios. Its allocation falls when the estimated risk premium is low or estimated volatility is high, which can reduce exposure during some market downturns. Nevertheless, the strategy remains exposed to model risk and can incur substantial losses when rolling estimates are inaccurate or market conditions change abruptly.

A rigorous performance comparison should report annualized return, annualized volatility, Sharpe ratio, maximum drawdown, turnover, and results net of transaction costs. Statistical or bootstrap uncertainty around these metrics would also help distinguish persistent performance from sample-specific outcomes.

6.7 Limitations

The empirical exercise has several important limitations.

1. **Expected-return estimation error.** Historical mean returns are noisy, particularly over a 12-month window. Because the Merton weight is highly sensitive to the estimated risk premium, estimation error can lead to unstable allocations.

2. **Constant risk-free rate.** The backtest fixes the annual risk-free rate at 4%, although actual short-term rates varied substantially over the sample. A time-varying Treasury bill series would provide a more realistic financing and cash-return assumption.

3. **Risk-aversion calibration.** The choice of γ is not estimated from investor behavior and materially affects the recommended portfolio. Results should be reported across a range of risk-aversion values.

4. **Transaction costs and turnover.** Monthly rebalancing is assumed to be costless. Bid–ask spreads, commissions, taxes, and market impact would reduce realized performance.

5. **Model assumptions.** Constant drift and volatility, normally distributed shocks, continuous trading, and frictionless borrowing are strong simplifications. Real markets exhibit jumps, volatility clustering, regime changes, and borrowing constraints.

6. **Backtest design.** The exercise should guard against look-ahead bias, survivorship bias, inconsistent return definitions, and parameter choices selected after observing the sample.

7 Conclusion and Future Work

This paper develops the classical Merton portfolio problem under CRRA utility. The HJB approach yields a closed-form risky allocation,

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2},$$

which increases with the market risk premium and decreases with volatility and risk aversion. Under the model's assumptions, the optimal risky fraction is constant over time, although terminal-wealth risk grows with the investment horizon.

The simulation exercise illustrates the central trade-off in the model: more aggressive allocations can increase average terminal wealth while also increasing dispersion and downside risk. The certainty-equivalent analysis shows that expected-return estimation error creates a welfare loss that is quadratic in the error and accumulates over time. The historical backtest suggests that a constrained Merton rule can adapt market exposure and perform competitively under selected assumptions, but the result is sensitive to parameter estimation, constraints, and implementation choices.

Several extensions would make the analysis more realistic. Intermediate consumption could be incorporated directly into the optimization problem. The risk-free rate, expected return, and volatility could be allowed to vary over time, including stochastic-volatility or regime-switching specifications. Portfolio constraints, transaction costs, taxes, and distinct borrowing and lending rates could be modeled explicitly. Robust or Bayesian portfolio methods could reduce sensitivity to expected-return estimates. Finally, the empirical analysis could be expanded to multiple risky assets, longer estimation windows, alternative markets, and a full set of out-of-sample risk and performance statistics.

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